Streaming catalogue and activity database

Group 6 | PostgreSQL 18 + pgvector | Physical foreign keys shown with solid lines

users | 24 rows

user_id [PK]	integer NOT NULL
full_name	text NOT NULL
email [UNIQUE]	text NOT NULL
country	char(2) NOT NULL
plan	text NOT NULL
preferences	jsonb NOT NULL

GIN(preferences). Required favourite_genres array; optional preference keys.

1 — 0..* — users.user_id

activity_logs | 144 rows

activity_id [PK]	bigint NOT NULL
user_id [FK]	integer NOT NULL
title_id [FK]	integer NULL
occurred_at	timestamptz NOT NULL
event_data	jsonb NOT NULL

GIN(event_data). Types: play, pause, rate, search. Search has no title; other types require a title.

0..* activities

0..1 title

search_phrases | 5 rows

phrase_id [PK]	integer NOT NULL
phrase	text NOT NULL
embedding	vector(384) NOT NULL

No foreign key cosine comparison

titles | 60 rows

title_id [PK]	integer NOT NULL
title	text NOT NULL
kind	text NOT NULL
release_year	integer NULL
genre	text NULL
description	text NOT NULL
embedding	vector(384) NULL

HNSW(embedding vector_cosine_ops). All 60 supplied title embeddings are populated.

JSONB examples (actual sample payload shapes)

```
preferences:
{"favourite_genres":["Mystery","Sci-fi"],
 "subtitle_language":"en"}
event_data (selected fields):
{"event_type":"play","minutes_watched":21}
{"event_type":"pause","paused_at_minute":11}
{"event_type":"rate","stars":2}
{"event_type":"search",
 "query":"a cozy mystery in a small town"}
```

PK = primary key; FK = foreign key. All users and events are fictional. AI-assisted draft; review and adapt before submission.